Martin Restrepo

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Summary

Systems and Computing Engineer moving from large-scale cloud infrastructure into robotics and autonomous systems. I build end-to-end perception-to-actuation pipelines on embedded hardware: camera calibration, fiducial-based pose estimation, object detection and inverse kinematics. 3+ years of software engineering experience, including high-availability distributed systems that serve some of the world's largest airlines. Seeking an M.S. in Autonomous Systems, with a focus on guidance, navigation and control of aerial systems.

Education

B.S. **Universidad de los Andes**, Systems and Computing Engineering, Minor in Economics — Bogotá, Colombia
- GPA 4.1 / 5.0 — 2019 – 2023

Experience

Amadeus, Cloud Engineer — Bogotá, Colombia
Nov 2025 – present
1 year 1 month
- Deploy and operate high-throughput services on Kubernetes that handle traffic for some of the world's largest airlines, under a 99.9% availability commitment.
- Designed and built Python bots for Microsoft Teams that alert the team in real time when new tickets arrive and auto-assign them round-robin across the day's on-shift engineers, cutting response times and balancing workload.
- Developed custom OpenShift plugins that automate recurring operational tasks, reducing manual effort and the risk of human error.
- Designed and implemented real-time monitoring and observability in Grafana and Splunk for proactive system-health management.
- Provide Tier-3 support and rapid incident response for mission-critical aviation software, consistently meeting strict SLAs.

CaseWare, Full-Stack Software Developer — Bogotá, Colombia
Jan 2024 – Sept 2025
1 year 9 months
- Reduced API response times by 40% by building and optimizing Spring Boot microservices on AWS.
- Cut production issues by 25% by introducing automated testing (Jest, Cypress, Mockito).
- Optimized the Imports application that ingests general ledgers and trial balances from Excel, CSV, Xero and QuickBooks.
- Built CI/CD workflows in GitHub Actions that run test suites and report results to a central monitoring dashboard.

Freelance, Mobile Application Developer — Bogotá, Colombia
Aug 2022 – Aug 2023
1 year 1 month
- Developed and published two mobile apps (Flutter, React Native) on the App Store and Google Play, with 2,500+ downloads.
- Built their backends with Express.js and AWS Amplify.

Universidad de los Andes, Teaching Assistant — Bogotá, Colombia
Jan 2020 – June 2020
6 months
- Taught Algorithms and Object-Oriented Programming II to 25 students.

Projects

EBA_01, Vision-Guided Robotic Arm
Independent project, ongoing.

- Built the perception pipeline for an ESP32-CAM video stream: ChArUco camera calibration, ArUco marker detection and pose estimation, and an affine pixel-to-table mapping from three reference markers.
- Derived and implemented inverse kinematics in C++ for a 3-DOF arm with a gripper on an ESP32-C3, decoupled into base rotation plus a 2-link planar solve, and fed it the vision output through a Wi-Fi HTTP endpoint.
- Fabricated and assembled 3D-printed parts; handled hardware integration, servo debugging and firmware setup (PlatformIO).
- Trained CNN image classifiers in Keras 3 with a PyTorch backend.
- Stack: Python, OpenCV, C++, ESP32-CAM, ESP32-C3, Arduino, PlatformIO.

Quadrotor GNC in ROS 2, Custom EKF and Trajectory-Tracking Controller
Independent project, in progress (started Nov 2026).

- Building a ROS 2 (Humble) stack that flies a simulated quadrotor in PX4 SITL / Gazebo through offboard control.
- Implementing a cascaded PID position controller for trajectory tracking, with an LQR controller planned for comparison.
- Implementing an Extended Kalman Filter that fuses IMU data with GPS and AprilTag-based visual pose, evaluated against ground truth, including GPS-dropout scenarios.
- Stack: ROS 2, PX4, Gazebo, Python / C++.

Awards

1st place, Systems and Computing Engineering projects, LA MUESTRA (Innovation Week) — Universidad de los Andes, Faculty of Engineering, Nov 2022
LiteContract, a web app that explains employment contracts in plain language.

2nd place, ExpoAndes engineering and innovation contest — Universidad de los Andes, Faculty of Engineering, 2019
LINASALUD team project.

Certificates

Drone Applications in Agribusiness (in progress, 25 h) — Dec 2026

Python for Computer Vision with OpenCV and Deep Learning

SOLIDWORKS Mastery

Complete Linux Training Course

Networking

Skills

Robotics & Vision

Embedded & Design

Software

Cloud & Systems

Languages

Spanish
Native

English
C1 (IELTS Academic 7.0, Feb 2026)